

Module 5: Data and AI Model Governance

100 Tutorial Topics: Data Governance, Model Risk Management, Validation, GenAI Governance

GARP RAI Exam Preparation

T1: Introduction to Model Governance

TOPIC: What is Model Governance?

DEFINITION:

- Sound governance of models is essential for effective and prudent risk management
- Without it, a range of outcomes and decisions may occur that can prove detrimental
- Model governance should exist across the entire model lifecycle
- From model development through performance monitoring and ultimately decommissioning

WHY GOVERNANCE MATTERS FOR AI/ML:

- Ensures technological advances don't overshadow need for fundamental challenges
- Addresses opacity of AI/ML models
- Proper governance of data used to train these models is particularly relevant
- Generative AI poses additional governance challenges

FOCUS:

- Financial sector
- Quantitative risk models (QRMs)
- Established formal regulatory framework around QRMs
- Principles applicable to wide range of industries

EXAM TRAP: Governance must span **the entire model lifecycle** from development to decommissioning!

T2: Data Governance Definition

TOPIC: Data Governance - Definition and Scope

DATA GOVERNANCE INSTITUTE DEFINITION:

- "A system of decision rights and accountabilities for information-related processes"
- "Executed according to agreed-upon models which describe who can take what actions"
- "With what information, and when, under what circumstances, using what methods"

BROAD SCOPE:

- Encompasses the people, processes, and technologies required to manage and protect data assets
- Covers entire data lifecycle from collection to deletion
- Balances data accessibility with data protection

KEY COMPONENTS:

- Data strategy and vision
- Data quality management
- Data provenance and legal compliance
- Data classification and security
- Metadata management

EXAM TRAP: Data governance is about **decision rights and accountabilities** for data!

T3: Data Strategy - Vision and Goals

TOPIC: Developing a Data Strategy

KEY REQUIREMENTS:

- Clear and approved data-governance policy
- Strategic vision and operational practices
- Level of authorization for non-publicly available personal information
- Specification of data usage (single or recurring use)
- Documentation of data sources

ALTERNATIVE DATA CONSIDERATIONS:

- Additional controls and testing required
- Privacy concerns
- Data quality and accuracy risks
- Unbounded data streams

DATA GOVERNANCE COMMITTEE RESPONSIBILITIES:

- Data collection, reconciliations, treatment, access, monitoring
- Management of data stewards
- Overall oversight on all aspects of data governance

EXAM TRAP: Data strategy requires **clear policies** and **authorization** for data use!

T4: Data Quality - Accuracy, Consistency, Integrity

TOPIC: Ensuring Data Quality

CORE REQUIREMENTS:

- Complete, relevant, transparent, and accurate data should be the primary objective
- Model risk management requirements around data quality do not change with AI/ML algorithms

ALTERNATIVE DATA SOURCES - ADDITIONAL SCRUTINY:

- Verifying accuracy of alternative data sources may be difficult
- Data providers may change field definitions after deployment
- Even well-established data sources are subject to this risk

EXAMPLE - COVID-19 PANDEMIC:

- U.S. Bureau of Labor Statistics unemployment data was biased
- Inability to conduct proper surveys
- Anomalies may occur with any data stream
- New data sources introduce additional risks

DATA MONITORING:

- Input data monitoring to alert model owners of flawed inputs
- Outlier detection may be insufficient to spot anomalies
- Definitional changes may only be disclosed in footnotes

EXAM TRAP: Data quality is **critical and requires continuous monitoring!**

TOPIC: Data Provenance - Verifying Legal Right to Use Data

DEFINITION:

- Provenance refers to the sequence of ownership and handling of items of value
- Now applies to data used to create models
- Proving legal right to use data is essential

WHY IT MATTERS:

- Scrutiny of whether vendors have legal right to use data
- Copyright and intellectual property lawsuits against GenAI vendors
- Example: Data scraped from websites without permission

REGULATORY CONSEQUENCES:

- FTC: Any model built on improperly obtained data must be destroyed
- First such legal action has already been settled with model destruction
- No model can be approved without proving ownership of input data
- Vendor models need to provide proof and indemnify licensees

KEY CHALLENGE:

- Established data (e.g., FICO) is generally safe
- Alternative data requires scrutiny
- Risk must be considered for GenAI models where model risk management applies

EXAM TRAP: Data provenance is **essential for legal compliance and model approval!**

T6: Data Classification

TOPIC: Data Classification - Structure and Confidentiality

DATA TYPES:

1 Quantitative Data:

- Measured or determined numerically
- Examples: income, age, temperature

2 Qualitative Data:

- Determined non-numerically
- Examples: gender, ethnicity, job title

DATA SOURCES:

- Observed data
- Projected data
- Judgment-based data

CRITICAL GOVERNANCE REQUIREMENT:

- Distinguish the classification of data being used
- Ensure access and use is permissible
- When data is confidential or personally identifiable, controls must be:
 - Present
 - Operational
 - Effective

EXAM TRAP: Data classification determines [access controls](#) and [usage permissions](#)!

T7: Metadata Management

TOPIC: Metadata Management

WHAT IS METADATA?

- Basic information about data
- Includes: file type, time of creation, data author
- Data source, file size, and other relevant information

TYPES OF METADATA:

- 1 **Descriptive:** Identifies and describes data
- 2 **Structural:** How data is organized
- 3 **Administrative:** Technical and rights information
- 4 **Reference:** Standards and definitions
- 5 **Statistical:** Data quality and methodology

ROBUST METADATA MANAGEMENT STRATEGY:

- Data documentation
- Data mapping
- Data dictionary
- Data definitions
- Data process flows
- Data relationships to other data
- Data structures

EXAM TRAP: Metadata management enables **better-informed business decisions!**

T8: Data Security and Protection

TOPIC: Data Security and Protection

KEY CONCEPTS:

- Data security often used interchangeably with data protection and data integrity
- Strategic security process and data protection plan are essential
- Procedural steps identified to safeguard:
 - Availability of data
 - Access to data
 - Data privacy

DATA PROTECTION STRATEGY:

- Safeguarding important information from:
 - Corruption
 - Malicious or accidental damage
 - Compromise or loss
- Importance increases with amount of data created and stored
- Data retention policy should be in place and adhered to

US REGULATORY EXAMPLES:

- Safeguards Rule (FTC): Financial institutions must keep customer information secure
- Gramm-Leach-Bliley Act (GLBA): Privacy practices and opt-out rights
- Privacy Act of 1974: Federal agency data collection and use

EXAM TRAP: Data security requires [comprehensive protection strategies and regulatory compliance!](#)

TOPIC: Data Access - Permissions and Secure Sharing

MOST IMPORTANT ASPECT:

- Ensuring data is readily accessible and shareable
- To those who have requisite permission and need to access it

COMPONENTS OF DATA ACCESS:

- Data security
- Data protection
- Data permissions
- Data access controls

REGULATORY REQUIREMENTS:

- Organizations must implement appropriate security measures
- Encryption to safeguard data
- Ensure confidentiality and integrity
- Maintain compliance

EXAM TRAP: Data access balances [availability with security and permissions](#)!

TOPIC: Compliance - Legal and Regulatory Requirements

THREE BASIC PRINCIPLES:

1 Obtaining Consent:

- Data subjects must consent to data collection
- Informed consent is essential
- Consent can be withdrawn

2 Data Minimization:

- Collect only necessary data
- Most effective privacy protection
- Reduce risk and exposure

3 Ensuring Rights of Data Subjects:

- Right to access data
- Right to correct data
- Right to delete data
- Right to be forgotten

KEY REGULATORY FRAMEWORKS:

- GDPR (Europe)
- CCPA (California)
- PIPL (China)
- GLBA (US Financial)

EXAM TRAP: Data compliance centers on [consent](#), [minimization](#), and [data subject rights](#)!

T11: Board Responsibilities in Data Governance

TOPIC: Board of Directors' Role in Data Governance

CRUCIAL RESPONSIBILITIES:

1 Approval of Framework:

- Overall data governance framework and policies
- Clear guidelines on data quality, privacy, security
- Compliance with relevant regulations

2 Oversight of Compliance and Risk Management:

- Compliance with legal, regulatory, and ethical standards
- Data protection laws (GDPR), industry standards, internal policies

3 Risk Assessment and Management:

- Data breaches
- Data quality issues
- Misuse of data

4 Regular Review:

- Adapt to changing business needs
- New technologies
- Regulatory requirements

EXAM TRAP: Board sets the "tone at the top" for data governance culture!

TOPIC: Synthetic Data and AI Readiness

SYNTHETIC DATA USAGE:

- Increasing use to augment available data
- Mitigates data risks (client or personally identifiable data)
- Can be designed to avoid biases

ADVANTAGES OF SYNTHETIC DATA:

- No privacy risks (doesn't come from real data subjects)
- Can be more easily rid of problematic biases by design
- Scalable and customizable

DISADVANTAGES:

- May not be as precise as real data (especially visual data)
- May differ from real data in ways not obvious
- Requires careful validation

AI-READY DATA:

- Accurate, unbiased, well-structured
- Cleaned and suitable for training
- Improves accuracy and efficiency of AI models

EXAM TRAP: Synthetic data and AI readiness [require consideration in data strategy!](#)

T13: Data Governance Process

TOPIC: The Data Governance Process

KEY STEPS:

- 1 Data Strategy Development:
 - Vision and goals
 - Policies and procedures
 - Authorization levels
- 2 Data Collection:
 - Identify sources
 - Verify provenance
 - Ensure legal compliance
- 3 Data Quality Management:
 - Accuracy checks
 - Consistency validation
 - Integrity verification
- 4 Data Classification:
 - Structure and confidentiality
 - Access controls
- 5 Metadata Management:
 - Documentation
 - Data dictionary
 - Data mapping
- 6 Data Security and Compliance:
 - Protection measures
 - Regulatory compliance
 - Monitoring and auditing

EXAM TRAP: Data governance is a continuous process across the data lifecycle!

T14: Roles and Responsibilities in Data Governance

TOPIC: Key Roles in Data Governance

BOARD OF DIRECTORS:

- Approve framework and policies
- Oversee compliance and risk management
- Ensure regular review and updates

DATA GOVERNANCE COMMITTEE:

- Overall oversight on all aspects
- Approve data practices and policies
- Not responsible for operational aspects

DATA STEWARDS:

- Day-to-day data management
- Ensure data quality and consistency
- Implement policies

DATA OWNERS:

- Accountability for specific data assets
- Define access permissions
- Ensure proper use

EXAM TRAP: Clear roles and responsibilities are **essential for effective data governance!**

TOPIC: Data Governance - Key Takeaways

CORE COMPONENTS:

1 Strategy:

- Vision and goals
- Policies and authorization
- Alternative data controls

2 Quality:

- Accuracy, consistency, integrity
- Continuous monitoring
- Alternative data scrutiny

3 Provenance:

- Legal right to use data
- Copyright compliance
- Vendor verification

4 Classification:

- Structure and confidentiality
- Access controls

5 Security:

- Protection measures
- Regulatory compliance
- Data retention

EXAM TRAP: Data governance is the **foundation for responsible AI!**

T16: What is a Model?

TOPIC: Defining a Model (SR 11-7)

SR 11-7 DEFINITION:

- "The term 'model' refers to a quantitative method, system, or approach"
- "That applies statistical, economic, financial, or mathematical theories"
- "Techniques, and assumptions to process input data into quantitative estimates"

KEY ELEMENTS:

1 Quantity of Interest:

- Numerical object whose future value is uncertain
- Examples: portfolio value, revenue projection, client count

2 Potential Future Scenarios:

- Possible values of the quantity of interest
- Each scenario assigned a weight (probability)

3 Risk Measure:

- Summarizes essential information
- Examples: VaR, expected shortfall, average outcome

EXAM TRAP: A model is a **quantitative method for processing data into estimates!**

TOPIC: Three Main Elements of Quantitative Risk Modeling

1. QUANTITY OF INTEREST:

- Numerical object whose future value is uncertain
- Examples:
 - Portfolio value in ten business days
 - Revenue projection over next five years
 - Expected number of new clients by year-end

2. POTENTIAL FUTURE SCENARIOS:

- Possible values of the quantity of interest
- Each scenario assigned a weight (probability)
- Examples:
 - Portfolio value conditioned on investment decisions
 - Revenue after business model changes
 - New clients after advertising campaign

3. RISK MEASURE:

- Summarizes essential information from scenarios
- Examples:
 - Value at Risk (VaR)
 - Expected Shortfall (ES)
 - Average outcome across all scenarios

EXAM TRAP: Risk models provide **structured approach to envision the future!**

TOPIC: Key Considerations in Risk Modeling

1. COMPLETENESS OF SCENARIO SETS:

- Challenging to anticipate every potential future scenario
- Historical data may not reflect rare events
- Expert judgment can be difficult

2. FEEDBACK EFFECTS:

- Scenarios and decisions may influence other market participants
- Scenario sets and weights may need ongoing updates

3. COMMUNICATION OF RESULTS:

- Reports should provide summary of main assumptions
- Ensure transparency and avoid complacency
- Reflect perceived risk based on uncertainty and exposure

4. REGULATORY CONTEXT:

- SR 11-7 (2011) established regulatory standards
- OCC Handbook (2021) includes AI/ML perspectives

EXAM TRAP: Effective risk modeling requires [completeness](#), [awareness of feedback](#), and [clear communication](#)!

TOPIC: Model Governance Policies - Key Elements

CORE POLICY AREAS:

- ① **Defining What Constitutes a Model:**
 - Distinction between model and end-user tool
 - Gray areas need consistent adherence
 - Presence of uncertainty may indicate a model
- ② **Model Risk Definition:**
 - Agree on internal definition
 - Align with regulatory requirements
- ③ **Risk Tier Identification:**
 - Drives depth and frequency of validation
 - Materiality, complexity, exposure
- ④ **Model Validation Definition:**
 - Establish expectations for validation process
 - Documentation standards
- ⑤ **Model Risk Appetite:**
 - Document institution's risk appetite
 - Outline procedures for unacceptable risks

EXAM TRAP: Written policies are the **foundation of model governance frameworks!**

TOPIC: Model Documentation - SR 11-7 Guidance

KEY REQUIREMENTS:

- Strong governance includes documentation of model development and validation
- Sufficiently detailed to allow parties unfamiliar with a model to understand:
 - How the model operates
 - Limitations of the model
 - Key assumptions of the model

DOCUMENTATION BEST PRACTICES:

- 1 Comprehensive:
 - Cover all aspects of model lifecycle
 - Include development and validation
 - Document all key decisions
- 2 Accessible:
 - Clear language
 - Well-organized
 - Executive summaries
- 3 Maintained:
 - Updated with changes
 - Version control
 - Historical records

EXAM TRAP: Documentation must be sufficiently detailed for others to understand the model!

TOPIC: Model Inventory - Comprehensive Tracking

WHAT TO INCLUDE:

1 Model Identification:

- Model owner, user, and developer information
- Purpose and brief description
- Applicable products

2 Methodology:

- Description of model methodology
- Model classification (e.g., Monte Carlo, logistic regression)
- Whether it's an AI/ML model

3 Development:

- In-house or vendor developed
- Use and frequency
- Restrictions and limitations

4 Assumptions and Limitations:

- Key assumptions
- Known weaknesses
- Management overlays

5 Documentation and History:

- Locations of documentation and validation reports
- Access to model code and databases
- History of updates, approvals, validations

EXAM TRAP: Model inventory enhances **transparency and tracking** of all models!

TOPIC: Model Landscape - Understanding Interdependencies

WHY MODEL LANDSCAPE MATTERS:

- Models should not be viewed in isolation
- Outputs from one model often serve as inputs for others
- Transparency regarding interdependencies is vital

KEY CONSIDERATIONS:

1 Interdependencies:

- Upstream and downstream dependencies
- Impact on key business decisions
- Critical models identification

2 Aggregate Risk:

- Influenced by interdependencies
- Shared assumptions and methodologies
- Models that can affect multiple outcomes

3 Duplication:

- Similar models used in different parts
- Different assumptions
- Inefficiency and risk

EXAM TRAP: Model landscape shows **how models interact and affect each other!**

TOPIC: Registering AI/ML Models - New Aspects

1. COMPLEXITY OF METHODOLOGY AND DESIGN:

- AI/ML models learn autonomously in developer-defined range
- Consider: chosen methodologies (neural networks vs. linear regression)
- Interpretability indicators (number of hidden layers, parameters)

2. DATA USAGE:

- Volume of required data or number of features
- Complexity of data structures (unlabeled, metadata)
- Data quality (poorly labeled, low quality, unstructured)
- Variable interactions and transformations

3. OUTPUT PARAMETERS:

- Supervised vs. unsupervised learning
- Prediction vs. sentiment analysis vs. recommendations
- No direct way to evaluate output accuracy for unsupervised

4. MODEL RECALIBRATION:

- Static vs. continuous recalibration
- Retraining vs. complete redevelopment
- Initial structure maintained or not

EXAM TRAP: AI/ML registration requires **additional considerations beyond traditional models!**

TOPIC: Model Risk Ranking Factors

KEY FACTORS:

1 Materiality:

- Economic consequences of misuse
- Operational impact
- Reputational consequences
- Usage by different parties
- Impact on decisions, financial statements, or regulatory reporting

2 Complexity:

- May elevate risk ranking
- Even if materiality is low
- AI/ML models often have high complexity

3 Exposure:

- If model affects published financial reports
- Risk ranking may be elevated
- Even simple models can have high exposure

IMPORTANT NOTE:

- Minimum standards of model risk assessment should apply to all models
- Regardless of materiality
- Absence of assessment introduces its own model risk

EXAM TRAP: Risk ranking considers **materiality**, **complexity**, and **exposure**!

T25: Model Performance Monitoring Metrics

TOPIC: Performance Monitoring Metrics

CLASSIFICATION MODELS:

- Accuracy
- Precision
- Recall
- F1 Score
- Area Under the ROC Curve (AUC)
- Confusion Matrices

REGRESSION MODELS:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Coefficient of Determination (R-squared)
- Root Mean Square Error (RMSE)

ADDITIONAL METRICS:

- Clustering metrics
- Deep learning metrics
- GenAI performance metrics

KEY POINT:

- Performance monitoring must be tailored to model type
- Metric must be appropriate for the model
- Model validation should assess this

EXAM TRAP: Performance metrics must be [appropriate for the specific model type!](#)

TOPIC: Three Lines of Defense in Model Risk Management

FIRST LINE (Business and Model Owners/Developers):

- Owns the model and its performance
- Responsible for designing and implementing models
- Operates daily/weekly performance dashboards
- Owns data quality for model inputs
- First to react to threshold breaches

SECOND LINE (Independent Risk Management/Model Validation):

- Sets model risk management policy
- Conducts independent model validations
- Challenges First Line's performance assessments
- Maintains enterprise-wide model inventory

THIRD LINE (Internal Audit):

- Provides independent assurance to the board
- Reviews whether monitoring is appropriately archived
- Ensures policies are consistently followed

EXAM TRAP: Three lines provide **checks and balances in model risk management!**

TOPIC: First Line of Defense - Detailed Responsibilities

MODEL OWNERS:

- Accountable for proper development, validation, approval
- Findings remediation, implementation, use
- Ongoing performance monitoring and reporting
- Change management
- Capture model characterization in inventory
- Assess model materiality

MODEL DEVELOPERS:

- Central to creation of models
- Decisions throughout modeling cycle affect risk
- Proper documentation saves time
- Act as both producers and recipients of validation feedback

AI/ML CONSIDERATIONS:

- Ensure continuous learning models don't drift into unvalidated states
- Work closely with MLOps teams
- Data scientists must interpret metrics in business context

EXAM TRAP: Model owners and developers are **responsible for model performance and accountability!**

T28: Second Line - Model Risk Management

TOPIC: Second Line of Defense - Model Risk Function

KEY RESPONSIBILITIES:

① Framework Development:

- Develop and maintain model risk management framework
- Establish general validation standards

② Model Inventory:

- Maintain model inventory
- Combined with material aspects of model landscape

③ Materiality Thresholds:

- Set and maintain thresholds for model materiality
- Related to business decisions

④ Validation Activities:

- Ensure risk model validation activities occur
- Report validation results to board

AI/ML CONSIDERATIONS:

- Deep expertise in validating complex ML models
- Assessing novel performance metrics
- Data drift detection for high-dimensional data
- Fairness/bias assessments
- Tools to "look inside the black box"

EXAM TRAP: Second Line provides **independent challenge and oversight** of models!

TOPIC: Third Line of Defense - Internal Audit

KEY RESPONSIBILITIES:

1 Independent Assurance:

- Provide assurance to the board
- Model risk management framework is well-designed
- Operating effectively

2 Annual Reviews:

- Monitoring evidence appropriately archived
- Performance exceptions escalated according to policy
- Policies consistently followed and up-to-date

AI/ML CONSIDERATIONS:

- Audit teams need sufficient understanding of ML concepts
- Focus on robustness of controls over automated ML pipelines
- Data governance for ML training data
- Adequacy of Second Line challenge for novel ML approaches

KEY POINT:

- Audit is "meta-validation" or "validation of validation"
- Not the primary model risk management function
- Vital and necessary independent check

EXAM TRAP: Internal Audit provides **independent verification** of the entire MRM framework!

TOPIC: Communication and Interaction in Model Governance

KEY AREAS:

① Breach Notification:

- How does First Line formally communicate performance breaches?
- What information must be provided?

② Independent Review:

- Second Line's responsibilities and timelines
- Verifying issues and assessing materiality
- Challenging root cause analysis and remediation plans

③ Dispute Resolution:

- If First and Second Lines disagree
- Documented escalation path
- Model Risk Committee or Chief Risk Officer

④ Reporting:

- How findings flow to senior management and board?
- Consolidated and understandable manner

COMMON FAILURE POINTS:

- First Line incentivized to downplay poor performance
- Second Line lacking business context or technical depth
- Third Line focusing on process rather than substance

EXAM TRAP: Effective communication is **vital for successful model governance!**

TOPIC: Model Development Process - Overview

KEY STEPS:

- 1 Define Objectives and Scope:
 - Clear problem or risk being addressed
 - Applicable products
 - Required data and desired outcomes
- 2 Data Collection and Preprocessing:
 - Gather relevant data
 - Clean and transform data
 - Handle missing values or outliers
- 3 Data Analysis and Visualization:
 - Gain insights into data
 - Identify patterns or relationships
 - Understand limitations or biases
- 4 Feature Engineering:
 - Select and engineer appropriate features
 - Transform or combine variables
 - Dimensionality reduction

EXAM TRAP: Model development is a **systematic process** from objectives to implementation!

TOPIC: Model Development Process (Continued)

5. MODEL SELECTION:

- Choose modeling approach
- Comprehensive evaluation of alternatives
- Document rationale

6. MODEL TRAINING/CALIBRATION:

- Optimize parameters and hyperparameters
- Best fit data and minimize errors
- Avoid data leakage (using future or concurrent data)
- Use cross-validation

7. DEVELOPER TESTING:

- Thorough testing before deployment
- Evaluate performance
- Identify issues early

8. DOCUMENTATION:

- Clear statement of purpose
- Sound design, theory, and logic
- Robust methodologies
- Rigorous data quality assessment

EXAM TRAP: SR 11-7 requires [sound development process and thorough testing!](#)

TOPIC: SR 11-7 Guidance on Model Development

KEY REQUIREMENTS:

- Clear statement of purpose to ensure model is developed in line with its intended use
- Sound design, theory, and logic underlying the model
- Robust model methodologies and processing components
- Rigorous assessment of data quality and relevance
- Appropriate documentation

TESTING REQUIREMENTS:

- Integral part of model development
- Various components and overall functioning evaluated
- Show model is performing as intended
- Demonstrate accuracy, robustness, and stability
- Evaluate limitations and assumptions

IMPORTANT NOTE:

- Organizations should ensure development of judgmental and qualitative aspects is also sound

EXAM TRAP: SR 11-7 requires [comprehensive development and testing of all model aspects!](#)

T34: Software Testing Types

TOPIC: Types of Software Testing

TESTING CHRONOLOGICAL ORDER:

① Unit Tests:

- Validate each piece of code performs as designed

② Component Tests:

- Verify functionality of individual code sections

③ Integration Tests:

- Verify interfaces between components

④ System Tests:

- Ensure completely integrated system meets requirements

⑤ Performance Tests:

- Assess speed, responsiveness, and stability

⑥ Regression Tests:

- Verify system continues to function after modifications

⑦ Acceptance Tests:

- Determine if model is ready for deployment

⑧ Adversarial Testing:

- "Red team" evaluation of AI models
- Extreme scenarios and edge cases
- Prompt injection and unexpected queries

EXAM TRAP: Testing should be **comprehensive** and **start as early as possible**!

T35: White-Box vs Black-Box Testing

TOPIC: Testing Approaches

WHITE-BOX TESTING:

- Testers have access to internal data structures, algorithms, and code
- Testing at unit level from developer's perspective
- Line-by-line proofreading of code
- More effective for finding specific issues

BLACK-BOX TESTING:

- Treats software as a closed box
- From user's perspective
- No knowledge of internal implementation
- Reduces likelihood of bias

GRAY-BOX TESTING:

- Between white-box and black-box
- Informed by knowledge about functional specifications
- Knowledge of architecture of underlying code
- Does not access the code itself

WHEN TO USE:

- White-box: internally developed models, early development
- Black-box: proprietary vendor models, functional testing
- Gray-box: distributed applications, testing component integration

EXAM TRAP: Testing approach depends on **model type, development stage, and accessibility!**

TOPIC: Effective Test Configurations

KEY PRINCIPLES:

- Tests need to be appropriately configured
- Usefulness hinges on adequacy and sufficiency of configurations
- Testers should not limit configurations solely to artificial cases
- Most valuable test configurations emerge from real-world situations

LESS REALISTIC PARAMETERS (BETTER TESTING):

- Missing trades
- Excessive trades
- Incorrect scales or mappings
- Extreme parameter values
- Parameters estimated from insufficient data
- Data provided by inexperienced users
- Variations in compilers or hardware

RECOMMENDATION:

- Establish preliminary process that automatically generates test results from available data

EXAM TRAP: The less realistic the test parameters, the better the testing!

TOPIC: The Use Test - Qualitative Assessment

WHAT IS THE USE TEST?

- Examines model within its context
- Considers human interaction and actual usage
- Assesses acceptance of the model
- Evaluates interpretation and application of results

KEY FOCUS AREAS:

1 People, Not Numbers:

- Focuses on credible stories of acceptance
- Success and failure stories
- Technical challenges take a back seat

2 Crucial Questions:

- Are results utilized appropriately?
- Do they prove useful?
- If not, why?

3 Exploration Areas:

- Acceptance and trust
- Comprehensibility of results
- Communication of feedback to modelers

EXAM TRAP: Use test focuses on **how people interact with and use the model!**

T38: Use Test Reporting

TOPIC: Use Test - Reporting and Impact

REPORTING:

- Results typically presented to senior management
- Not documented as technical reports or spreadsheets
- Easier to communicate results
- Minimizes risk of being seen as arduous requirement

ORGANIZATIONAL IMPACT:

- Contributes to improving culture of risk modeling
- Enhances risk management culture
- Bridges gap between technical and business perspectives

BENEFITS:

- Promotes acceptance and trust
- Identifies usability issues
- Improves model communication
- Drives continuous improvement

KEY POINT:

- Use test is qualitative in nature
- Difficult to follow a standard procedure
- Requires ongoing commitment

EXAM TRAP: Use test reporting focuses on [senior management communication and culture improvement!](#)

TOPIC: Should There Be Limits on Model Use?

WHY LIMITS ARE NEEDED:

- Every QRM is based on underlying assumptions
- Assumptions should be clearly stated
- Identify model limitations and weaknesses
- Prevent misuse

TYPES OF LIMITS:

1 Product Applicability:

- Model approved for specific products only
- Example: Standard interest rate swaptions only
- Not for Bermudan swaption without additional validation

2 Access Restrictions:

- Who can use the model
- Training requirements
- Authorization levels

3 Portfolio Size Limits:

- Particularly for new markets or instruments
- Limited data or experience
- Risk ranking can change as portfolio grows

EXAM TRAP: Model limits prevent misuse and ensure risk ranking stays appropriate!

TOPIC: Three General Guidelines for Risk Modeling Culture

1. AWARENESS:

- Be aware of limitations and assumptions of risk modeling
- Understand your company's history with QRMs
- Know risks entailed and validation processes in place
- Stay informed about market practices
- Recognize that the world is constantly changing

2. TRANSPARENCY:

- Transparently communicate assumptions, limitations, and documentation
- Provide detailed documentation with executive summaries
- Document decision-making process
- Document all validation activities, including unsuccessful attempts
- Engage in open communication with end users

3. EXPERIENCE:

- Learn from past modeling endeavors
- Apply relevant lessons
- Emphasize proper project management
- Develop prototypes early
- Establish and maintain libraries of reusable code
- Seek input from other modelers and external experts

EXAM TRAP: Guidelines foster **culture that views validation as opportunity for insight!**

TOPIC: Adapting SR 11-7 Guidelines

KEY PRINCIPLE:

- SR 11-7 established regulatory guidance
- Each institution needs to adapt to specific situation

FACTORS TO CONSIDER:

1 Specific Models:

- Type of models used
- Complexity and risk profile

2 Institution Size and Structure:

- Large vs. small institution
- Organizational structure
- Resources available

3 Governance Framework:

- Existing governance structures
- Board and committee oversight

4 Regulatory Requirements:

- Specific regulatory expectations
- Jurisdictional requirements

5 Budget Constraints:

- Resources allocated to model risk
- Cost-benefit considerations

EXAM TRAP: SR 11-7 is general guidance that **must be adapted to each institution's situation!**

T42: Model Changes Process

TOPIC: Model Changes - Governance Process

1. CHANGE MANAGEMENT PROCESS:

- Structured process for making changes
- Steps: change requests, impact assessment, documentation, validation, approval

2. DOCUMENTATION:

- Comprehensive documentation required
- Reason for change, modifications, validation results, approvals

3. INDEPENDENT REVIEW:

- Ongoing validation of changes
- Performed by individuals not involved in modification
- Assess accuracy, reliability, compliance, and risks

4. APPROVAL:

- Final approval through defined governance process
- Detailed documentation in model inventory
- Rationale, decision, conditions, or limitations

5. REVIEW OF PROPOSED CHANGES:

- Model owner communicates change to MRM team
- Assessment for materiality
- Change-based validation if material

EXAM TRAP: Model changes require [structured governance and independent review!](#)

TOPIC: AI/ML Review Framework Differences

1. FREQUENCY:

- More frequent monitoring for AI/ML models
- Higher complexity with more parameters
- Undesirable outcomes under certain conditions
- Greater susceptibility to data drift
- Monitoring dashboards may be useful

2. VALIDATION STAKEHOLDERS:

- Wider range of stakeholders affected
- Data protection regulation and ethical policies
- Include HR, compliance, operational risk
- Approval structure modifications needed
- Higher familiarity with AI/ML necessary

3. VALIDATION CONTENT:

- Focus on explainability
- Benchmarking against traditional models
- Assessment based on different data sets
- Assessment of specific cases (edge cases)
- Back testing and cross-validation
- Reporting component for bias detection

EXAM TRAP: AI/ML models require **more frequent review and broader stakeholder involvement!**

TOPIC: Implementation and Monitoring of Model Changes

IMPLEMENTATION:

- Approved changes implemented in controlled manner
- Proper documentation
- Include changes to performance monitoring

MONITORING:

- Ongoing monitoring continues after implementation
- Ensure changes achieve intended objectives
- Remain compliant with regulatory requirements

AI/ML AUTOMATION:

- Growing volume of AI/ML models
- Potential need for more frequent retraining
- Greater impact of model errors
- Automated ongoing controls become more relevant

KEY POINT:

- Automated controls where practical and responsible
- Balance automation with oversight

EXAM TRAP: Model changes must be **implemented with controls and continuously monitored!**

TOPIC: Model Changes - Key Takeaways

KEY PRINCIPLES:

- ① **Structured Process:**
 - Change requests, assessment, validation, approval
- ② **Documentation:**
 - Comprehensive record of all changes
 - Rationale and approvals
- ③ **Independent Review:**
 - Ongoing validation
 - Objective assessment
- ④ **Frequency:**
 - AI/ML models need more frequent review
 - Address data drift and complexity
- ⑤ **Stakeholders:**
 - Broader range of stakeholders
 - Include compliance, HR, operational risk
- ⑥ **Monitoring:**
 - Continuous monitoring after changes
 - Automated controls where practical

EXAM TRAP: Model changes require **comprehensive governance throughout the lifecycle!**

TOPIC: What is Model Validation?

DEFINITION:

- Process of rigorously testing the model, inputs, outputs, governance, etc.
- To try to identify the extent of residual model risk
- Assess mitigating controls

KEY ASPECTS:

- Critical component in identifying model risk
- Involves qualitative and quantitative aspects
- Follows model lifecycle

WHEN VALIDATION OCCURS:

1 Initial Validation:

- Prior to implementation and usage
- Baseline assessment

2 Routine Periodic Validation:

- Annual reviews
- In-depth periodic baseline revalidations

3 Change-Based Validation:

- Triggered by material changes
- When model owner makes modifications

EXAM TRAP: Model validation is the rigorous testing to identify residual risk!

T47: Validation Lifecycle

TOPIC: Model Validation Lifecycle

KEY STAGES:

- 1 Model Identification:
 - Recognize and classify the model
- 2 Inventory:
 - Record in model inventory
 - Schedule for validation
- 3 Initial Validation:
 - Baseline validation prior to implementation
- 4 Production and Monitoring:
 - Back testing and performance monitoring
 - Ongoing throughout production usage
- 5 Periodic Validation:
 - Routine validations
 - Annual or more frequent
- 6 Change-Based Validation:
 - Triggered by material changes
- 7 Retirement:
 - Store in retired model inventory
 - Decommission

EXAM TRAP: Validation follows the [entire model lifecycle](#) from identification to retirement!

TOPIC: Model Validation Activities

KEY REVIEW AREAS:

1 Data Assessment:

- Appropriateness of data sources
- Data analysis (proxies, time series lengths)

2 Assumptions and Parameters:

- Assessment of any model assumptions
- Parameter evaluation

3 Limitations:

- Identification of limitations, strengths, weaknesses
- Approved use cases and users

4 Implementation:

- Appropriateness of implementation platform
- Usage of expert judgment and overlays

5 Conceptual Soundness:

- Review of conceptual soundness
- Assessment of benchmarks and best practices

6 Performance Monitoring:

- Analysis of ongoing monitoring plans
- Frequency, key risk indicators, escalation plans

EXAM TRAP: Validation covers **all aspects of the model and its use!**

TOPIC: Model Validation Activities (Continued)

ADDITIONAL REVIEW AREAS:

1 Documentation:

- Review quality of documentation
- Ensure backup plans exist (internal and external)

2 Controls:

- Review controls around implementation and usage
- Access controls

3 Outcomes Analysis:

- Appropriateness of dependent variable
- Support for business objectives

4 Testing:

- Stress and scenario tests
- Model boundary testing
- Parameter stability

5 Backtesting:

- Appropriate cadence with model usage
- VaR models: daily monitoring
- Credit risk models: monthly
- Asset-liability models: quarterly

EXAM TRAP: Validation includes ongoing testing and monitoring appropriate to model type!

T50: Initial Validation

TOPIC: Initial Validation - Three Key Goals

GOAL 1: OPERATIONAL FEASIBILITY:

- Checked and confirmed
- Model can run as intended without technical malfunctions
- Preproduction phase to run on intended schedule

GOAL 2: PROPER DOCUMENTATION:

- Adheres to firm-wide standards
- Model development standards
- Documentation standards
- Implementation standards
- Model monitoring standards
- Third-party governance standards
- Includes executive summaries

GOAL 3: USER TRAINING:

- Model users receive sufficient training
- Interpret and utilize results effectively

KEY POINT:

- Results form basis for official approval (sign-off)
- Assumptions must be justified
- Limitations and weaknesses documented

EXAM TRAP: Initial validation ensures **operational feasibility, documentation, and user training!**

T51: Ongoing Validation

TOPIC: Ongoing (Periodic) Validation

PRIMARY OBJECTIVE:

- Observe whether model remains aligned with intended purpose
- Assumptions remain valid
- Data still appropriate
- Performance monitoring indicates expected performance

KEY ACTIVITIES:

1 Methodology Reassessment:

- Still in line with best practices
- Reflects real world

2 Assumption Validation:

- Original assumptions remain valid
- Real world consistently challenges assumptions

3 Benchmark/Challenger Models:

- Support validation
- Compare results

IMPORTANT CONSIDERATIONS:

- Well documented
- Consolidated within single document
- Regular validation with fixed agenda
- Depth and frequency commensurate with risk tier

EXAM TRAP: Ongoing validation ensures **models remain fit for purpose over time!**

TOPIC: General Issues in Model Validation

1. TERMINOLOGY:

- Alternative terms: model review, model QA, model evaluation, model vetting

2. FOCUS ON SUITABILITY:

- Ultimate goal: assess if model is appropriate and effectively used
- Initial validation critical for credibility
- Ongoing performance monitoring as proposed and approved
- Validation must remain within intended application domain

3. RECURSIVE VALIDATION:

- Validation activities should be subject to critical examination
- Ensure validation doesn't rely on defective data
- Must remain within intended application domain
- Need for challenge of validation activities

4. NO "VALID" MODEL:

- Models are simplified representations under assumed conditions
- Goal is not to test for a "valid" model
- Subject model to attempts to invalidate it
- Successful validation = withstood rigorous testing

EXAM TRAP: Validation is about **testing**, not **proving validity**!

TOPIC: Usefulness vs. Validity

KEY PRINCIPLE:

- Models do not have to be perfect to be useful
- If weaknesses and limitations are identified:
 - Insights can improve the model
 - Restrict application
 - Enhance understanding of capabilities and limitations

IMPORTANT CONSIDERATIONS:

- Model weaknesses can arise from regime shifts
- Need for ongoing monitoring and validation over time

EFFECTIVE CHALLENGE (SR 11-7):

- Critical analysis by objective and informed parties
- Identify crucial model assumptions and limitations
- Spot and communicate relevant model weaknesses

REQUIREMENTS FOR EFFECTIVE CHALLENGE:

- Independence from model development
- High level of competence
- Sufficient degree of influence

EXAM TRAP: Models don't need to be perfect to be **useful**, but **limitations must be understood!**

TOPIC: Validation of GenAI Models - Special Considerations

KEY CHALLENGES:

1 Open-Ended Output:

- Unstructured input and output
- Makes validation challenging

2 Performance Metrics:

- Different and more complex than traditional AI/ML
- New metrics needed

3 Data Drift and Coverage:

- Responses may become obsolete
- Changing environment
- Model obsolescence

4 Hallucination:

- Tendency to generate factually incorrect responses
- Appear very convincing

5 Discrimination and Bias:

- Ensure responses are not discriminatory or biased
- AI can amplify bias in data

EXAM TRAP: GenAI validation requires [addressing unique challenges of generative models!](#)

T55: GenAI Validation Challenges (Continued)

TOPIC: GenAI Validation Challenges (Continued)

6. MULTI-MODAL ISSUES:

- Processing images, video, and audio data
- In addition to text
- More challenging to validate

7. INTERPRETABILITY AND EXPLAINABILITY:

- Results difficult to explain or interpret
- Black-box nature
- Regulatory compliance challenges

8. RAPID TECHNOLOGICAL CHANGE:

- More frequent review required
- Regular recalibration needed
- Keeping pace with developments

9. THIRD-PARTY AND OPEN-SOURCE MODELS:

- Increasingly offered as models as a service
- Particular attention needed
- Limited transparency

10. COMPUTATIONAL RESOURCES:

- Significant resources required
- Training, validation, recalibration
- Infrastructure challenges

EXAM TRAP: Human review is [essential in GenAI validation!](#)

T56: Model Design Motivation

TOPIC: Model Design - Motivation for Development

CHANGING CONDITIONS:

- New markets
- New products
- Customer requests
- Models must be validated

REGULATORY PRESSURE:

- Regulatory findings
- Model not performing as intended
- Redevelopment or replacement needed

RECENT CRISIS:

- Losses highlight urgency
- Enhanced or more sensitive modeling needed
- Lessons from crisis

INTERNAL CONCERNS:

- Senior management discomfort
- Changes in business model
- Increased exposure to innovative products
- Changing market conditions

INNOVATION AND GROWTH:

- New or evolving customer demands
- New products or model extensions
- Business acquisitions

EXAM TRAP: Model development is driven by **business needs, regulation, and changing conditions!**

T57: Model Developers

TOPIC: Model Developers - The "Quants"

REQUIRED EXPERTISE:

- Mathematics
- Econometrics
- Statistics
- Strong product knowledge

BACKGROUND:

- Advanced academic degrees
- Mathematics, physics, engineering, computer science
- Scientific background

KEY QUALITIES:

- Creative problem-solving
- Skepticism
- Out-of-the-box thinking
- Experience including past crises

IMPORTANT BALANCE:

- Most challenging aspect: designing and constructing potential future scenarios
- Assigning weights to scenarios
- Must be mindful of difficulties in risk definition
- Avoid uncritical use of distributions
- Critical reflection on assumptions

EXAM TRAP: Model developers need [quantitative expertise and practical experience!](#)

TOPIC: Modeling Issues - Discretization

WHAT IS DISCRETIZATION?

- Describing the world using continuous variables
- Cases where variables are discretized to finite values
- Reducing complexity in modeling path-dependent events
- Events occurring at discrete time intervals

WHY DISCRETIZE?

- Complex equations without closed-form solutions
- Numerical approximation via discretization
- Should yield very close results when properly implemented
- Time intervals divided into days, months, years

CHALLENGES:

- Discretization error
- Stability
- Convergence
- Computational complexity
- Boundary conditions
- Numerical precision
- Choice of time step is crucial

EXAM TRAP: Discretization enables implementation but **requires careful parameter selection!**

TOPIC: Modeling Issues - Approximations

WHAT ARE APPROXIMATIONS?

- Replacing complex or hard-to-evaluate components
- Alternative methods that produce similar results
- More conveniently implemented

KEY CONSIDERATIONS:

- Choosing an approximation fixes the precision
- No room for improvement except by selecting different approximation

EXAMPLES:

- Price sensitivities
- Taylor series approximation in market risk
- Valid for simple instruments and small changes

TIME HORIZON SCALING:

- Scaling VaR by square root of time
- Assumes normal distribution and temporal independence
- Often convenient but rarely empirically justified
- Validity must be examined via statistical tests

EXAM TRAP: Approximations simplify but **fix precision and require careful validation!**

T60: Modeling Issues - Numerical Evaluation

TOPIC: Modeling Issues - Numerical Evaluation

WHAT IS NUMERICAL EVALUATION?

- Evaluation of model component with potential for arbitrary precision
- Equipped with parameters controlling precision

METHODS:

- Computation of integrals
- Evaluating non-elementary functions
- Solving systems of equations
- Linear algebra problems
- Interpolation
- Numerical differential equation solutions
- Optimization

KEY FACTORS:

- **Stability:** Small errors should not lead to unbounded imprecision
- **Convergence:** Results approach true value as precision increases
- **Floating-Point Arithmetic:** Rounding errors may affect stability

TRADE-OFF:

- Running time vs. precision
- Theoretical confirmation rarely sufficient
- Extensive testing necessary

EXAM TRAP: Numerical evaluation requires **balancing precision with computational cost!**

TOPIC: Modeling Issues - Random Numbers

WHY RANDOM NUMBERS NEEDED:

- Models rely heavily on probability theory
- Assigning values to random variables
- Samples drawn from distributions

CHALLENGES:

- Generating random numbers is difficult for both humans and computers
- Computers use pseudo-random number generators (RNGs)
- Depend on initial seed
- Cannot produce actual random numbers

BEST PRACTICE:

- Rely on well-designed deterministic pseudo-RNGs
- Rather than introducing randomness in choice of RNG

RNG QUALITY ASSESSMENT:

- Specialized test suites
- Criteria: speed and reproducibility

EXAM TRAP: Random number generation requires [high-quality deterministic pseudo-RNGs](#)!

TOPIC: Model Implementation Tasks

1. MODEL DESIGN:

- Designed and documented by model designers
- Facilitate translation into computer code

2. CORE IMPLEMENTATION:

- Inner workings implemented as black box
- With specified interfaces

3. SYSTEM IMPLEMENTATION:

- Model core integrated into system
- Provides user interfaces
- Collects input data and parameters
- Schedules computations
- Feeds the core
- Processes output data
- Keeps history of computations

KEY CONSIDERATIONS:

- Different skills for core vs system implementation
- Different programming languages
- Specialized skills needed for system implementation

EXAM TRAP: Implementation involves **model design, core development, and system integration!**

TOPIC: Model Adaptation Tasks

WHY ADAPTATION NEEDED:

- No model lasts forever
- Periodic reviews detect weaknesses
- Demand adaptation or revision

ADAPTATION TYPES:

① Model Adaptation:

- Model and documentation adapted (or replaced)
- Existing computer code adapted

② Core Adaptation:

- Model core adapted (or replaced)
- Interfaces may be adapted

③ System Adaptation:

- System around model core adapted (or replaced)

CHALLENGES:

- Initial implementation team may have left
- External consultants may have done some tasks
- If model no longer understood, adaptation is challenging
- If same people, risk of ignorance or entrenched poor practices

EXAM TRAP: Adaptation requires **understanding the model and careful change management!**

T64: Running a Model

TOPIC: Running a Model - Key Considerations

ROBUSTNESS:

- Models can become complex
- Intertwining with multiple systems
- Increased risk of failure
- Minimize complexity
- Eliminate exotic systems
- Standardize interfaces
- Avoid "reinventing the wheel"

COMPUTATIONAL CONSIDERATIONS:

- Total response time varies greatly
- Credit risk models: may take weeks
- Market risk models: daily reporting
- Pre-deal checking: faster processing
- Overall running time determined by slowest task
- Over-optimization for specific hardware can be risky

FLEXIBILITY:

- Scheduled runs: regular numbers with minimal manual intervention
- Ad hoc runs: additional information on short notice
- Staff knowledgeable about model interfaces
- Prevent gradual shift from scheduled to ad hoc mode

EXAM TRAP: Running models requires **robustness, appropriate computing, and flexibility!**

T65: Model Adaptation and Running Summary

TOPIC: Implementation, Adaptation, and Running - Key Takeaways

IMPLEMENTATION:

- Model design, core implementation, system integration
- Different skills for different tasks
- Documentation is essential

ADAPTATION:

- Periodic reviews detect weaknesses
- Model, core, or system adaptation
- Challenges with changing teams

RUNNING:

- Robustness and standardization
- Computational considerations
- Scheduled vs. ad hoc runs

COMMON ISSUES:

- Over-optimism
- Inadequate budgeting
- Midstream redefinition of goals
- Misunderstandings and miscommunication
- Programming errors
- Lack of software quality assurance
- Inappropriate design architecture

EXAM TRAP: Model operations require careful management of implementation, adaptation, and running!

TOPIC: Potential Sources of Misinterpretation

1. LACK OF COMPREHENSION:

- Users lack comprehensive understanding of model
- Underlying assumptions unknown
- Limitations of outputs unclear
- Necessary knowledge and expertise missing

2. NEGLECTING MODEL LIMITATIONS:

- Models have inherent limitations
- May assume certain conditions
- May not fully capture all relevant risks
- Failing to acknowledge limitations leads to flawed understanding

3. OVEREMPHASIS ON POINT ESTIMATES:

- Models provide estimates with uncertainty
- Focusing solely on point estimates
- Ignoring range of potential outcomes
- Understanding probability distributions is crucial

EXAM TRAP: Misinterpretation can have significant implications for decision-making!

TOPIC: Potential Sources of Misinterpretation (Continued)

4. FAILURE TO CONSIDER QUALITATIVE FACTORS:

- Models focus on quantitative measures
- Qualitative factors should not be overlooked
- Market conditions, regulatory changes, external influences
- Exclusion can lead to distorted understanding

5. IGNORING DECISION CONTEXT:

- Model results considered independently
- Without specific context of decision
- Models should inform decision making within context
- Incorporate broader business strategies, risk appetite

6. CONFIRMATION BIAS:

- Users interpret results to align with preconceived beliefs
- Cloud judgment
- Compromise objective interpretation
- Approach results with open mind and commitment to unbiased analysis

EXAM TRAP: Misinterpretation arises from **cognitive biases** and **lack of context**!

T68: Fostering Proper Interpretation

TOPIC: Fostering Proper Interpretation of Model Results

KEY PRACTICES:

1 Training and Education:

- Ensure users understand model
- Explain assumptions and limitations
- Provide context for results

2 Clear Communication:

- Present results with uncertainty ranges
- Include confidence intervals
- Explain probabilistic interpretations

3 Open Dialogue:

- Encourage questioning of results
- Discuss alternative interpretations
- Foster critical thinking

4 Culture:

- Foster culture of open dialogue
- Encourage critical thinking
- Cross-functional collaboration

KEY POINT:

- Guard against misinterpretation biases
- Approach results with open mind
- Commit to unbiased analysis

EXAM TRAP: Proper interpretation requires **training, clear communication, and supportive culture!**

TOPIC: Recommendations for AI/ML Model Governance

1. BEGIN WITH EXISTING FRAMEWORKS:

- Don't start from scratch
- Use experience from traditional model validation
- Build on a strong foundation

2. CONSIDER THE NEW ROLE OF DATA:

- AI/ML is "model-free" – everything depends on data
- Address issues such as:
 - Various forms of bias
 - Overfitting
 - Population drift
 - Model drift and regime changes
 - Risk of harmful output
- New uses will reveal further challenges:
 - Data limitations
 - Biases
 - Privacy and right to use
 - Legality

EXAM TRAP: AI/ML governance must **address the central role of data!**

TOPIC: Recommendations for AI/ML Model Governance (Continued)

3. ADD NEW PERSPECTIVES:

- AI/ML increases relevance of ethical aspects
- Data bias
- Explainability of model results
- Recalibration process
- These attributes need to be considered in model inventory

4. BIS SURVEY FINDINGS - DATA GOVERNANCE:

- Adequate documentation
- Policies, standards, procedures
- Data quality - accuracy, transparency, provenance
- Security and responsible use
- Comprehensive data management systems
- Metadata on sources, version, curation
- Asset inventories and metadata registries

EXAM TRAP: AI/ML governance must incorporate ethical considerations and robust data governance!

TOPIC: BIS Survey - Organizational Structures for AI Governance

COMMON APPROACHES:

1 Centralized Solutions:

- Steering committees/commissions
- Program management offices
- Multi-modal structures

2 Hub-and-Spoke Systems:

- Overall network with multiple functional hubs
- Connected to various departments

3 Fully Decentralized:

- Governance at business line level
- Local control

ACCOUNTABILITY ASSIGNMENTS:

- Existing roles: Chief Data/Information/Technology Officer
- Existing units: IT department
- Newly created functions
- Informal setups: communities of practice, cross-functional networks

EXAM TRAP: AI governance structures range from **centralized to decentralized with various accountability models!**

TOPIC: BIS Survey - Guidelines and Risk Management

GUIDELINES AND POLICIES:

- Terms of use for AI systems
- Modalities, interfaces, and data
- Models, third-party dependencies
- Upstream/downstream data
- Guidelines for responsible AI use
- Privacy standards
- Explainability requirements
- International principles (e.g., Fundamental Principles of Official Statistics)
- Documentation of methods for regular assessment

RISK MANAGEMENT FRAMEWORKS:

- Regular audit of AI systems for vulnerabilities
- Evaluation of robustness of safety measures
- Incident response procedures
- Business continuity plans
- Processes to promote and maintain skills
- Prevent erosion of knowledge due to overreliance
- Leverage existing frameworks (three lines of defense)

EXAM TRAP: Comprehensive guidelines and risk management are **essential for AI governance!**

TOPIC: Why GenAI Governance Is Harder

1. OPACITY AND UNPREDICTABILITY:

- Functioning almost always opaque
- Output unpredictable
- Difficult to interpret or audit
- Probabilistic nature: same prompt may lead to different outputs
- Reproducibility challenges

2. EMERGENT CAPABILITIES:

- Behaviors not anticipated by developers
- As models become larger and more sophisticated
- Skills not observed in smaller models
- Changing capabilities hard to assess

3. POST-DEPLOYMENT SHIFTS:

- Model behavior can shift after deployment
- External providers may update or fine-tune models
- Changes in prompts or connected data alter performance
- One-off validation is inadequate

EXAM TRAP: GenAI governance must address **opacity, unpredictability, and evolving capabilities!**

TOPIC: Internal Governance Levers for GenAI

1. ASSIGN OWNERSHIP AND ACCOUNTABILITY:

- Clarify responsibility at each stage
- Tool selection, testing, deployment, monitoring
- Ensure responsibility doesn't fall between teams

2. RISK-BASED REVIEW AND DEPLOYMENT:

- Tiered review gates
- Full risk reviews for high-impact systems
- Light checks for low-stakes internal tools

3. INCENTIVIZE RESPONSIBLE BEHAVIOR:

- Risk-tied performance metrics
- Reward flagging risks, documenting limitations
- Align team incentives with responsible use

4. STAFF TRAINING:

- Equip staff with knowledge of GenAI risks
- Data protection, hallucinations, complaints

EXAM TRAP: Internal governance requires **clear ownership, risk-based reviews, and proper incentives!**

TOPIC: External Governance and Adaptive Tools

EXTERNAL GOVERNANCE CONSTRAINTS:

- Track evolving legal and regulatory requirements
- EU AI Act, Digital Services Act, US state laws
- Evaluate responsibilities with model providers
- Understand incident policies and update schedules
- Align with emerging standards (NIST AI RMF, ISO/IEC 42001)

ADAPTIVE TOOLS AND PRACTICES:

- Continuously monitor post-deployment usage
- Use red-teaming and adversarial testing
- Develop scenario plans for unpredictability
- Deploy technical safeguards (watermarking, usage logging)
- Enable feedback loops and update mechanisms

EXAM TRAP: GenAI governance requires [external awareness](#) and [adaptive practices](#)!

TOPIC: Strategic Risk Assessment Questions

KEY QUESTIONS:

- ❶ Is GenAI the right tool for this application?
 - Ensure reasons for deploying are good reasons
 - Know what trade-offs are to be expected
- ❷ What are the risks to:
 - Safety
 - Legality
 - Reputation
 - User trust
 - Transparency
- ❸ Risk Identification:
 - Identify relevant risks before deployment
 - Map potential unpredictable behaviors
 - Assess possible impacts

KEY POINT:

- Strategic assessment should occur before deployment
- Consider both intended and unintended consequences

EXAM TRAP: Strategic assessment ensures **GenAI is the right tool for the right application!**

TOPIC: Model Selection and Deployment Questions

KEY QUESTIONS:

- 1 What model is being used?
 - What do we know about its training data?
 - What do we know about its alignment?
- 2 Do we control the model?
 - API vs open source
 - Control over updates
 - Transparency varies

WHY THESE QUESTIONS MATTER:

- Understanding training data reveals potential biases
- Control determines ability to audit and modify
- Transparency affects governance capabilities

KEY POINT:

- Model selection affects all subsequent governance
- Choose models that align with governance capabilities

EXAM TRAP: Model selection determines **governance options and risk exposure!**

TOPIC: Data Governance Questions

KEY QUESTIONS:

1 What data is being used?

- Is any of it proprietary or sensitive?
- Legal restrictions (GDPR, etc.)
- Client confidentiality
- Intellectual property

2 Are data protection and retention policies in place?

- Trace how inputs are used by providers
- Understand data flow

WHY THESE QUESTIONS MATTER:

- Data provenance affects legal compliance
- Data protection policies prevent breaches
- Understanding data flow enables governance

KEY POINT:

- Data governance is foundational for GenAI
- Know what data is used and how it's protected

EXAM TRAP: Data governance questions are **essential** for responsible GenAI use!

TOPIC: Technical Risk Control Questions

KEY QUESTIONS:

- ① Have we tested for unpredictability?
 - Hallucinations
 - Model drift
 - Unexpected behaviors
- ② What safeguards are in place against misuse?
 - Guardrails
 - Human review triggers
 - Usage limits
 - Logs and monitoring

WHY THESE QUESTIONS MATTER:

- Unpredictability can lead to harmful outcomes
- Safeguards prevent misuse and limit damage
- Monitoring enables detection and response

KEY POINT:

- Technical controls are essential for safe deployment
- Test for edge cases and unexpected behaviors

EXAM TRAP: Technical risk controls are essential for safe GenAI deployment!

TOPIC: Organizational Structure and Incentive Questions

KEY QUESTIONS:

- ① Who owns risk for GenAI?
 - Is that person/team empowered?
 - Clear accountability
- ② Are staff appropriately trained?
 - Know of unpredictable behavior
 - Understand risks and limitations
 - Prevent unintended consequences
- ③ Are incentives aligned with responsible AI use?
 - Not just speed to deploy
 - Risk-aware behaviors rewarded

WHY THESE QUESTIONS MATTER:

- Ownership ensures accountability
- Training prevents misuse
- Incentives drive behavior

EXAM TRAP: Organizational structure must support responsible GenAI use!

T81: External Environment Questions

TOPIC: External Environment Questions

KEY QUESTIONS:

1 Are we prepared for changing laws and regulation?

- EU AI Act
- US state legislation
- Compliance frameworks

2 Are we aligned with best practices?

- NIST AI RMF
- ISO/IEC 42001
- Industry standards

WHY THESE QUESTIONS MATTER:

- Regulatory landscape is evolving
- Compliance is mandatory
- Best practices build trust

KEY POINT:

- Following recognized frameworks strengthens compliance
- Builds trust with external stakeholders
- Demonstrates commitment to responsible AI

EXAM TRAP: External alignment is **essential** for regulatory compliance and stakeholder trust!

T82: Questions for Risk Professionals Summary

TOPIC: Questions for Risk Professionals - Summary

STRATEGIC RISK:

- Is GenAI the right tool?
- What are the risks?

MODEL SELECTION:

- What model is being used?
- Do we control it?

DATA GOVERNANCE:

- What data is being used?
- Are protection policies in place?

TECHNICAL CONTROLS:

- Have we tested for unpredictability?
- What safeguards are in place?

ORGANIZATIONAL:

- Who owns risk?
- Are staff trained?
- Are incentives aligned?

EXTERNAL:

- Are we prepared for regulation?
- Are we aligned with best practices?

EXAM TRAP: A structured questioning approach [supports comprehensive GenAI governance!](#)

T83: Final Thoughts - Team Qualification

TOPIC: Final Thoughts - Team Qualification

VALIDATION TEAM REQUIREMENTS:

- Knowledge of AI/ML techniques
- Hands-on expertise
- Strong foundations in:
 - Product knowledge
 - Econometrics
 - Statistics
 - Computer science
- Proactive approach to staying abreast of latest advancements

ADDITIONAL CONSIDERATIONS:

- Augment validation staff
- Add AI/ML validation team
- Hands-on knowledge of:
 - AI/ML development platforms
 - Cloud-based environments
 - Low-code and no-code environments
 - CI/CD and MLOps platforms

KEY POINT:

- Familiarity with AI/ML technologies beyond traditional knowledge
- Specific training may be necessary
- Incorporate specialists
- Use external experts if needed

EXAM TRAP: Validation teams need [updated skills and expertise for AI/ML models!](#)

TOPIC: Final Thoughts - End-to-End Review

REVIEW FRAMEWORK:

- Entire model validation framework needs review
- Regulators consider AI/ML models within scope of model risk management
- SR 11-7, being principle-based, applies to AI/ML models

IMPLICATIONS:

- Challenges in model design
- Implications related to:
 - Data
 - Implementation
 - Monitoring
 - Documentation
 - Use

KEY POINT:

- AI/ML models fall within existing regulatory frameworks
- Comprehensive review is necessary

EXAM TRAP: End-to-end review is **essential** for AI/ML model validation!

T85: Final Thoughts - Balance and GenAI

TOPIC: Final Thoughts - Balance and GenAI

MODEL COMPLEXITY BALANCE:

- More complex is not always better
- Balance model performance with:
 - Interpretability
 - Feedback from supervisors
 - Cost and effort of implementation
 - Maintenance and monitoring
 - In-house expertise
 - Availability of reputable code libraries
 - Academic underpinning
- Validation teams need balanced approach:
 - Not just regulatory-driven
 - Not just performance-oriented
 - Nuanced cost-benefit analysis

GENAI GOVERNANCE:

- Requires particular focus
- Support development of shared standards
- Transparency around GenAI
- Longer-term credibility depends on:
 - Internal compliance
 - Broader governance ecosystem

EXAM TRAP: Balance and GenAI governance are [essential for sustainable model risk management!](#)

TOPIC: Key Points on Model Development and Testing

DEVELOPMENT STEPS:

- 1 Define objectives and scope
- 2 Data collection and preprocessing
- 3 Data analysis and visualization
- 4 Feature engineering
- 5 Model selection
- 6 Model training/calibration
- 7 Developer testing
- 8 Documentation

TESTING TYPES:

- Unit tests
- Component tests
- Integration tests
- System tests
- Performance tests
- Regression tests
- Acceptance tests
- Adversarial testing

EXAM TRAP: Model development and testing are **systematic processes requiring comprehensive approaches!**

T87: Model Validation - Key Points

TOPIC: Key Points on Model Validation

VALIDATION LIFECYCLE:

- 1 Model identification
- 2 Inventory
- 3 Initial validation
- 4 Production and monitoring
- 5 Periodic validation
- 6 Change-based validation
- 7 Retirement

VALIDATION ACTIVITIES:

- Data assessment
- Assumptions and parameters
- Limitations identification
- Implementation review
- Conceptual soundness
- Performance monitoring
- Documentation review
- Controls review
- Outcomes analysis
- Testing and backtesting

EXAM TRAP: Validation is a **comprehensive, ongoing process throughout the model lifecycle!**

T88: Model Misinterpretation - Key Points

TOPIC: Key Points on Misinterpretation

COMMON SOURCES:

- ❶ Lack of comprehension
- ❷ Neglecting model limitations
- ❸ Overemphasis on point estimates
- ❹ Failure to consider qualitative factors
- ❺ Ignoring decision context
- ❻ Confirmation bias

PREVENTION STRATEGIES:

- Training and education
- Clear communication
- Open dialogue
- Supportive culture

KEY POINT:

- Misinterpretation can have significant implications
- Requires active management
- Culture of critical thinking is essential

EXAM TRAP: Misinterpretation can be prevented through **education, communication, and culture!**

TOPIC: Three Lines of Defense - Key Points

FIRST LINE (Model Owners/Developers):

- Own the model and its performance
- Design and implement models
- Operate performance dashboards
- First to react to breaches

SECOND LINE (Risk Management/Validation):

- Set model risk management policy
- Conduct independent validations
- Challenge First Line assessments
- Maintain model inventory

THIRD LINE (Internal Audit):

- Provide independent assurance
- Review framework effectiveness
- Ensure policy compliance

EXAM TRAP: Three lines provide **checks and balances** in model risk management!

TOPIC: Model Changes - Key Points

CHANGE MANAGEMENT PROCESS:

- 1 Change requests
- 2 Impact assessment
- 3 Documentation
- 4 Validation
- 5 Approval
- 6 Implementation
- 7 Monitoring

AI/ML CONSIDERATIONS:

- More frequent monitoring needed
- Broader stakeholder involvement
- Focus on explainability
- Benchmarking against traditional models
- Assessment on different data sets
- Assessment of specific cases
- Back testing and cross-validation
- Reporting for bias detection

EXAM TRAP: AI/ML model changes require **more frequent review and broader stakeholder involvement!**

TOPIC: Data Governance - Key Points

CORE COMPONENTS:

1 Strategy:

- Vision and goals
- Policies and authorization

2 Quality:

- Accuracy, consistency, integrity
- Continuous monitoring

3 Provenance:

- Legal right to use data
- Copyright compliance

4 Classification:

- Structure and confidentiality
- Access controls

5 Security:

- Protection measures
- Regulatory compliance

EXAM TRAP: Data governance is the **foundation for responsible AI and model management!**

TOPIC: Model Inventory - Key Points

WHAT TO INCLUDE:

- 1 Model identification (owner, user, developer)
- 2 Purpose and description
- 3 Methodology and classification
- 4 Development source (in-house/vendor)
- 5 Assumptions and limitations
- 6 Documentation locations
- 7 History of updates and validations
- 8 Performance monitoring frequency
- 9 Regulatory use indication
- 10 Risk tier

AI/ML ADDITIONAL CONSIDERATIONS:

- Complexity of methodology
- Data usage and quality
- Output parameters
- Model recalibration needs
- Risk ranking factors
- Performance monitoring metrics

EXAM TRAP: Model inventory provides **comprehensive tracking and transparency** for all models!

TOPIC: Model Risk Ranking - Key Points

KEY FACTORS:

1 Materiality:

- Economic consequences
- Operational impact
- Reputational consequences
- Usage by different parties
- Impact on decisions, financial statements, regulatory reporting

2 Complexity:

- May elevate risk ranking
- Even if materiality is low

3 Exposure:

- Published financial reports
- Wide impact
- Regulatory significance

IMPORTANT NOTE:

- Minimum standards for all models regardless of materiality
- Absence of assessment introduces its own risk

EXAM TRAP: Risk ranking determines **validation frequency and intensity!**

TOPIC: Performance Monitoring - Key Points

CLASSIFICATION MODELS:

- Accuracy, Precision, Recall, F1, AUC
- Confusion Matrices

REGRESSION MODELS:

- MAE, MSE, R-squared, RMSE

ADDITIONAL METRICS:

- Clustering metrics
- Deep learning metrics
- GenAI performance metrics

KEY POINT:

- Performance monitoring must be tailored to model type
- Metrics must be appropriate for the model
- Model validation should assess metric appropriateness

EXAM TRAP: Performance metrics must be [appropriate for the specific model type](#)!

TOPIC: GenAI Validation - Key Points

SPECIAL CONSIDERATIONS:

- 1 Open-ended output
- 2 Different performance metrics
- 3 Data drift and coverage
- 4 Hallucination
- 5 Discrimination and bias
- 6 Multi-modal issues
- 7 Interpretability and explainability
- 8 Rapid technological change
- 9 Third-party and open-source models
- 10 Computational resources

ESSENTIAL:

- Human review is essential
- Regular validation and recalibration
- Particular attention to third-party models

EXAM TRAP: GenAI validation requires [addressing unique challenges and human review!](#)

TOPIC: GenAI Governance - Key Points

INTERNAL LEVERS:

- 1 Clear ownership and accountability
- 2 Risk-based review and deployment
- 3 Incentivize responsible behavior
- 4 Staff training
- 5 Validation and monitoring

EXTERNAL CONSTRAINTS:

- 1 Track legal and regulatory requirements
- 2 Evaluate responsibilities with providers
- 3 Align with emerging standards

ADAPTIVE TOOLS:

- 1 Continuous monitoring
- 2 Red-teaming and adversarial testing
- 3 Scenario planning
- 4 Technical safeguards
- 5 Feedback loops

EXAM TRAP: GenAI governance requires **internal**, **external**, and **adaptive approaches**!

T97: Questions for Risk Professionals - Summary

TOPIC: Questions for Risk Professionals - Summary

STRATEGIC RISK:

- Is GenAI the right tool?
- What are the risks?

MODEL SELECTION:

- What model is being used?
- Do we control it?

DATA GOVERNANCE:

- What data is being used?
- Are protection policies in place?

TECHNICAL CONTROLS:

- Have we tested for unpredictability?
- What safeguards are in place?

ORGANIZATIONAL:

- Who owns risk?
- Are staff trained?
- Are incentives aligned?

EXTERNAL:

- Are we prepared for regulation?
- Are we aligned with best practices?

EXAM TRAP: Structured questioning [supports comprehensive GenAI governance!](#)

TOPIC: Implementing AI Governance - BIS Survey Findings

DATA GOVERNANCE:

- Adequate documentation
- Data quality (accuracy, transparency, provenance)
- Security and responsible use
- Comprehensive data management systems
- Metadata on sources, version, curation
- Asset inventories and metadata registries

ORGANIZATIONAL STRUCTURES:

- Centralized solutions (steering committees, PMOs)
- Hub-and-spoke systems
- Decentralized approaches
- Accountability assignments

EXAM TRAP: BIS survey shows [diverse approaches to AI governance implementation!](#)

TOPIC: Implementing AI Governance - BIS Survey (Continued)

GUIDELINES AND POLICIES:

- Terms of use for AI systems
- Guidelines for responsible AI use
- Privacy standards
- Explainability requirements
- International principles
- Documentation of assessment methods

RISK MANAGEMENT FRAMEWORKS:

- Regular audit of AI systems
- Evaluation of safety measures
- Incident response procedures
- Business continuity plans
- Skills maintenance processes
- Leverage existing frameworks (3LoD)

EXAM TRAP: Comprehensive guidelines and risk management are [essential for AI governance!](#)

TOPIC: Data and AI Model Governance - Comprehensive Summary

KEY PILLARS:

- ① **Data Governance:**
 - Strategy, quality, provenance, classification, security
- ② **Model Governance:**
 - Policies, documentation, inventory, risk ranking
- ③ **Model Development:**
 - Objectives, data, features, training, testing
- ④ **Model Validation:**
 - Initial, ongoing, change-based
 - Comprehensive review activities
- ⑤ **Three Lines of Defense:**
 - Owners/developers, risk management, internal audit
- ⑥ **GenAI Governance:**
 - Internal levers, external constraints, adaptive tools

EXAM SUCCESS: Understand data governance, model risk management, validation, and GenAI governance!